

ACFI315 Finance with Excel

Lecture 2 Forecasting in Excel (Linear Relationship)

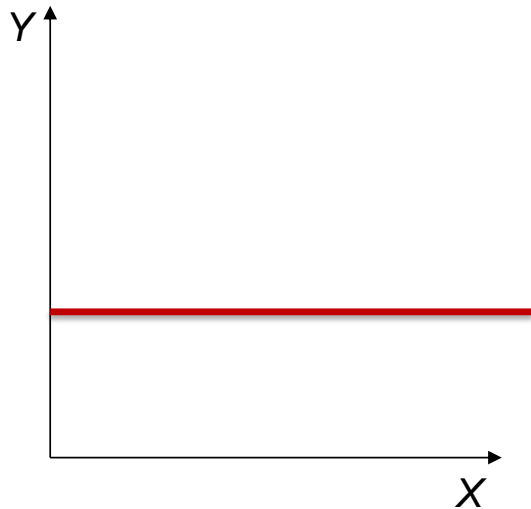


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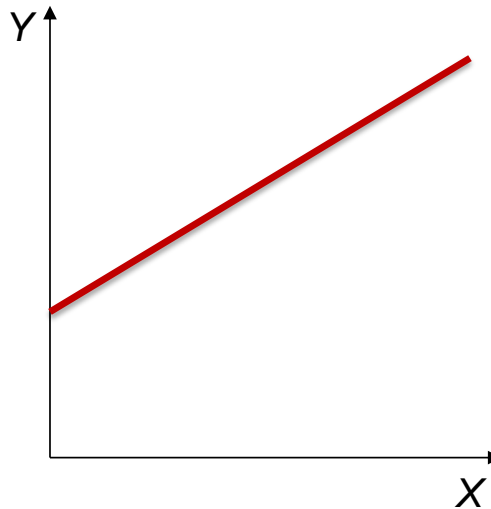
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Linear Relationship

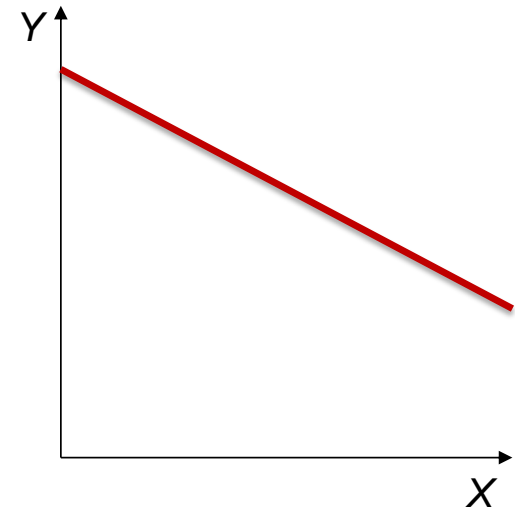
- Assume that we have the relationship $Y = a + bX$ with $a > 0$, could you graph the relationship in different situations (e.g., $b = 0$, $b > 0$, and $b < 0$)?



$b = 0$



$b > 0$



$b < 0$

Covariance

- Define the covariance between Y and X , which measures direction of association between two variables:

$$COV(\tilde{X}\tilde{Y}) = E[(\tilde{X} - E(\tilde{X}))(\tilde{Y} - E(\tilde{Y}))]$$

- How can you obtain an estimate of $COV(XY)$ using EXCEL?
 - We could use the **COVARIANCE.P** function in EXCEL.
 - **COVARIANCE.P** returns population covariance, the average of the products of deviations of each data point pair in two data sets.
 - We could use covariance to determine the relationship between two data sets.
- The syntax of COVARIANCE.P is COVARIANCE.P(array1,array2), which requires following inputs:
 - Array1 – Required. The first cell range of integers.
 - Array2 – Required. The second cell range of integers.

Correlation

- Define the correlation between Y and X , which measures the direction and strength of the association between variables:

$$\text{Corr}(XY) = \frac{\text{COV}(XY)}{\sigma_X \sigma_Y}$$

- The value of $\text{Corr}(XY)$ is between -1 and 1.
 - If $\text{Corr}(XY) = 1$, X and Y are perfectly positively correlated
 - If $\text{Corr}(XY) = -1$, X and Y are perfectly negatively correlated
 - If $\text{Corr}(XY) = 0$, X and Y are independent
- Correlation measures the direction and strength of the association between X and Y whereas covariance only measures direction.

Correlation Calculation in EXCEL

- We could use the **CORREL** function in EXCEL to calculate the correlation coefficient of two cell ranges.
- The syntax of **CORREL** is CORREL(array1,array2), which requires following inputs:
 - Array1 – Required. The first cell range of integers.
 - Array2 – Required. The second cell range of integers.

Coefficients Estimation in Excel

- Consider the model $Y = a + bX$, if we would like to obtain $\hat{a}$ and $\hat{b}$, the estimate of a and b , we could use two alternative ways.
- First, we could draw a plot of Y against X using EXCEL and then insert a trendline (In this case a linear trend). Then, we could format the trend line and ask EXCEL to display the equation which yields $\hat{a}$ and $\hat{b}$.
- Second, we could use **INTERCEPT** and **SLOPE** functions in EXCEL.

INTERCEPT Function in Excel

- We could obtain $\hat{a}$ using the **INTERCEPT** function in Excel.
 - The **INTERCEPT** function calculates the point at which a line will intersect the y-axis by using existing x-values and y-values.
 - The intercept point is based on a best-fit regression line plotted through the known x-values and known y-values.
 - Use the **INTERCEPT** function when you want to determine the value of the dependent variable when the independent variable is 0.
- The syntax of **INTERCEPT** is INTERCEPT(known y's,known x's), which requires following inputs:
 - Known y's – Required. The dependent set of observations or data.
 - Known x's – Required. The independent set of observations or data.

SLOPE Function in Excel

- We could obtain $\hat{b}$ using the **SLOPE** function in Excel.
 - The **SLOPE** function returns the slope of the linear regression line through data points in known y's and known x's.
 - The slope is the vertical distance divided by the horizontal distance between any two points on the line, which is the rate of change along the regression line.
- The syntax of **SLOPE** is SLOPE(known y's,known x's), which requires following inputs:
 - Known y's – Required. An array or cell range of numeric dependent data points.
 - Known x's – Required. The set of independent data points.

R-Square

- R-square (R^2) measures the proportion of variation in the dependent variable Y explained by the model. If $R^2 = 1$, the model is a perfect fit.
- For a linear model, the relationship between R^2 and $Corr(XY)$ is $R^2 = Corr(XY)^2$.
- There are two ways to obtain R^2 in EXCEL.
- First, once we draw a plot of Y against X using EXCEL and then insert a linear trendline, we could format the trend line and ask EXCEL to display the equation and the R^2 .
- Second, we could use the **RSQ** function in EXCEL.
 - The **RSQ** function returns the square of the Pearson product moment correlation coefficient through data points in known y's and known x's. The r-square value can be interpreted as the proportion of the variance in Y attributable to the variance of X .
- The syntax of **RSQ** is **RSQ(known y's,known x's)**, which requires following inputs:
 - Known y's – Required. An array or range of data points.
 - Known x's – Required. An array or range of data points.

Forecasting in Excel

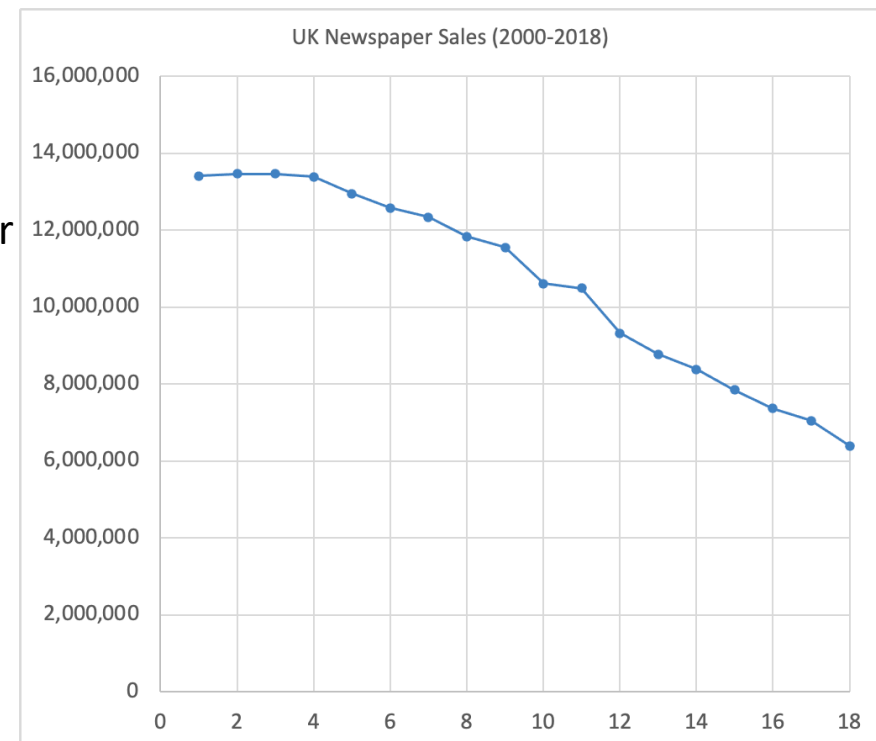
- Once we obtain coefficient estimation for the linear model in EXCEL, there are two alternative ways to obtain forecasts for future periods.
- First, once we draw a plot of Y against X using EXCEL and insert a linear trendline, we could format the trend line and ask EXCEL to provide a forecast for future periods.
- Second, we could use **FORECAST** function in EXCEL, telling EXCEL the future value of X .
 - The **FORECAST** function calculates, or predicts, a future value by using existing values. The future value is a y-value for a given x-value. The existing values are known x-values and y-values. The future value is predicted by using linear regression.
 - The syntax is FORECAST(x,known y's, known x's).
- Third, we could use the **TREND** function in EXCEL.
- Fourth, we could use INTERCEPT and SLOPE and the new x value for future periods to obtain a brute force forecast.

TREND Function in Excel

- The **TREND** function returns values along a linear trend. It fits a straight line (using the method of least squares) to the arrays known y's and known x's. TREND returns the y-values along that line for the array of new x's that you specify.
- The syntax is TREND(known y's, [known x's],[new x's],[const])
- Known y's – Required. The set of y-values you already know in the relationship $Y = a + bX$.
- Known x's – Optional. An optional set of x-values that you may already know in the relationship $Y = a + bX$.
- New x's – Optional. New x-values for which you want TREND to return corresponding y-values.
- Const – Optional. A logical value specifying whether to force the constant a to equal to 0.

Example

- Consider the circulation of UK newspaper from 2000 to 2018 depicted below. Imagine that you use this data to estimate $Sales_t = a + bTime_t$
- a. What is the expected sign of $\hat{a}$ and $\hat{b}$?
- b. Does the linear model appear suitable or adequate in your opinion? Please try and justify your view.



Answers

- a. What is the expected sign of $\hat{a}$ and $\hat{b}$?
 - $\hat{a} > 0$ and $\hat{b} < 0$. Sales of newspapers are positive but are declining over time.
- b. Does the linear model appear suitable or adequate in your opinion? Please try and justify your view.
- The answers to this question is open to interpretation as it is based on visual inspection of the diagram. If your eyeballs tell you that sales are declining linearly with time, the linear model may well be adequate.
- On the other hand, if you inspect the diagram and suspect that the decline is non-linear then a polynomial model (next week) in time could be more appropriate.